Shape memory alloy (Bending Wire)

Introduction

I have an experiment on shape memory alloy wire that is modeled with Lagoudas procedure via MATLAB SIMULINK , Testing the wire behavior due to the changing temperature and constant bending load ,then controlling it via PI controller ,then discussing the results and how a linear controller acts to nonlinear system as SMA wire and discussing how feedforward controller could be a solution to many problems that occur in PI controller approach .

1.Lagoudas shape memory effect parameters

Austenite Young's modulus (Ea)	70 GPa
Martensite Young's modulus(Em)	30 GPa
Maximum Transformation Strain (et)	.045
Austenite Starting Temperature (As)	280 K
Austenite finishing temperature (Af)	300 K
Martensite Starting Temperature (Ms)	275 K
Martensite Finishing Temperature (Mf)	256 K
Slope of Austenite Limit Curve (Ca)	8 MPa/K
Slope of Martensite Limit Curve (Cm)	8 MPa/K
Heat Capacity At Constant Pressure(Ka)	400 J/K
Density (Rho)	$6500\ Kg/m^3$
Diameter of The Wire (D)	.5 mm
Length of The Wire(L)	100 mm

2.The shape memory effect experiment

First I apply sin wave temperature input to the model at constant load with amplitude 70 and bias 255 and observe the input /output curve.

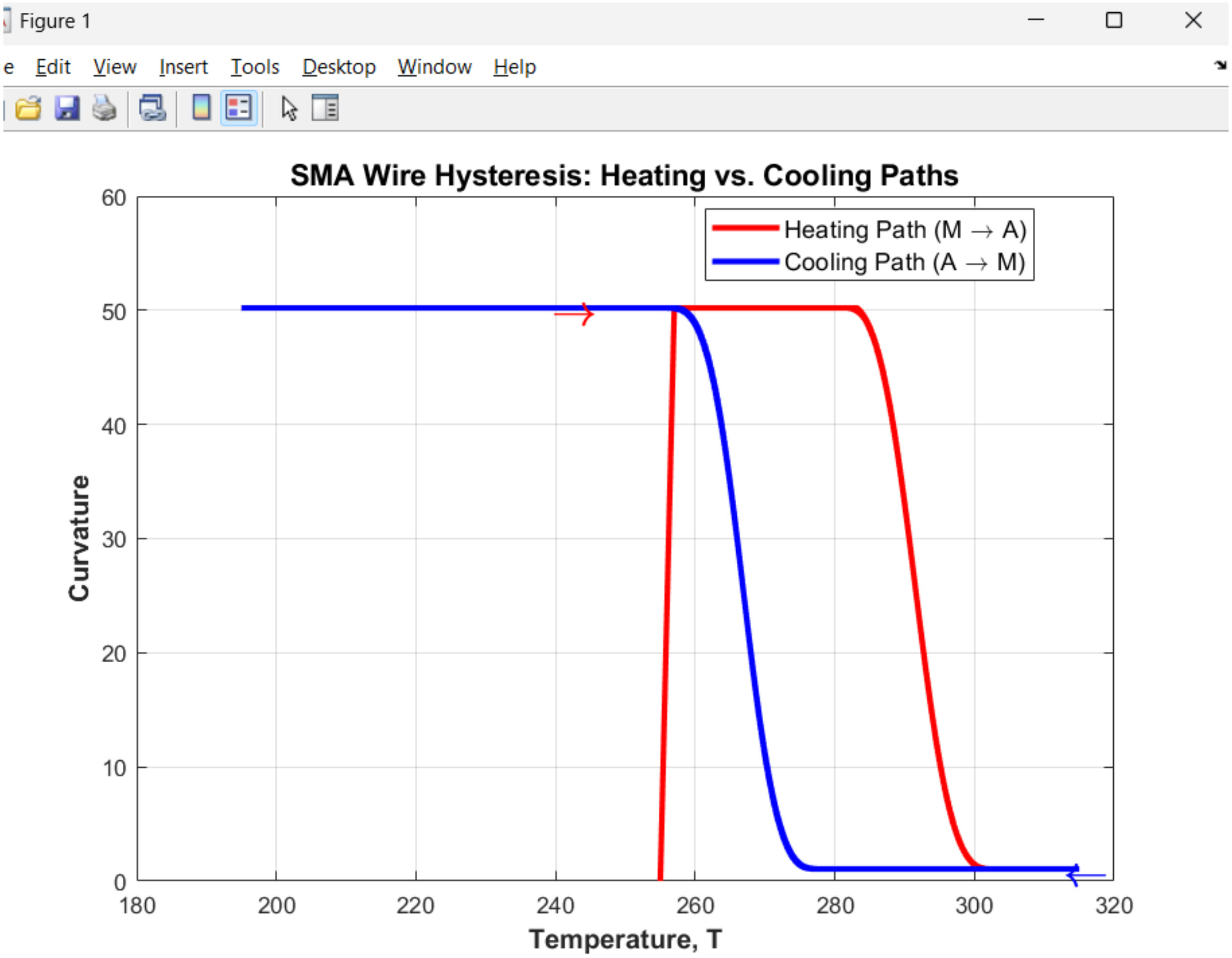

Figure (1)

At first instant of the simulation the bending load is applied on the tip of the wire is represented as the sudden jump to 50 then as the temperature rising from 255 K to nearly 280 K which is the austenite starting temperature the wire behavior never changes, From 280 K and above the wire starts to act and get closer to the initial shape From nearly 320 K to 275 K the wire behavior again never change and from 275 K to 256 K it loses its stiffness and starts to get affected due to the bending load again to end of the simulation at 185 K the figure (1) plot shows the hysteresis of the system the wire never acts instantly to the increasing or decreasing of the temperature

3.The linear negative gain PI position control

As shown before the wire is bent to 50 rad so the range we can control the wire within is from nearly 1 rad to 50 rad.

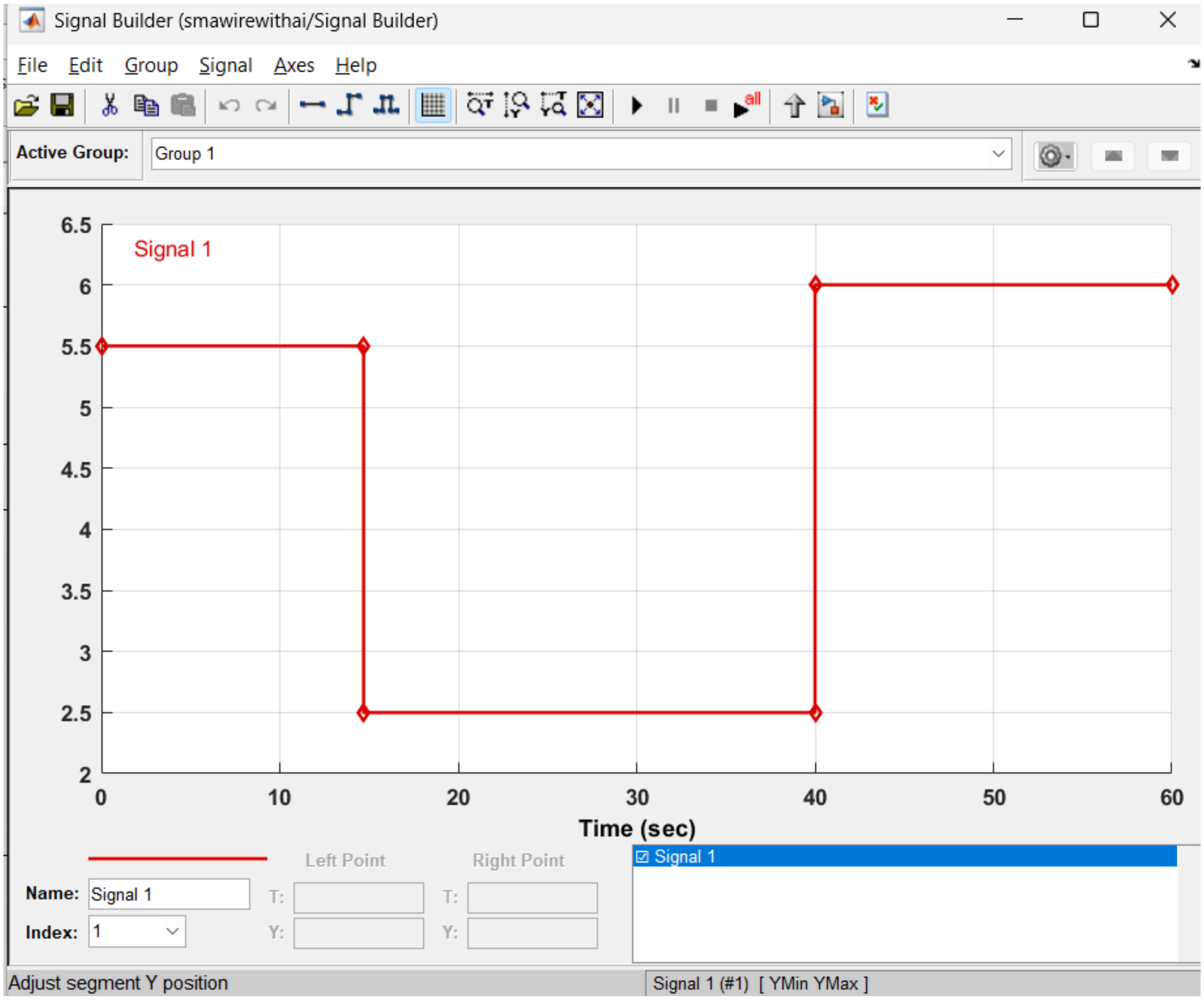

Figure (2)

The input signal as Figure (2) starts at 5.5 rad lasts 15 seconds, goes to 2.5 rad and lasts 25 seconds then goes to 6 rad to the end of the simulation, using a negative gain position controller as the wire contracts due to heat rising and submits to the load when the temperature cools down .

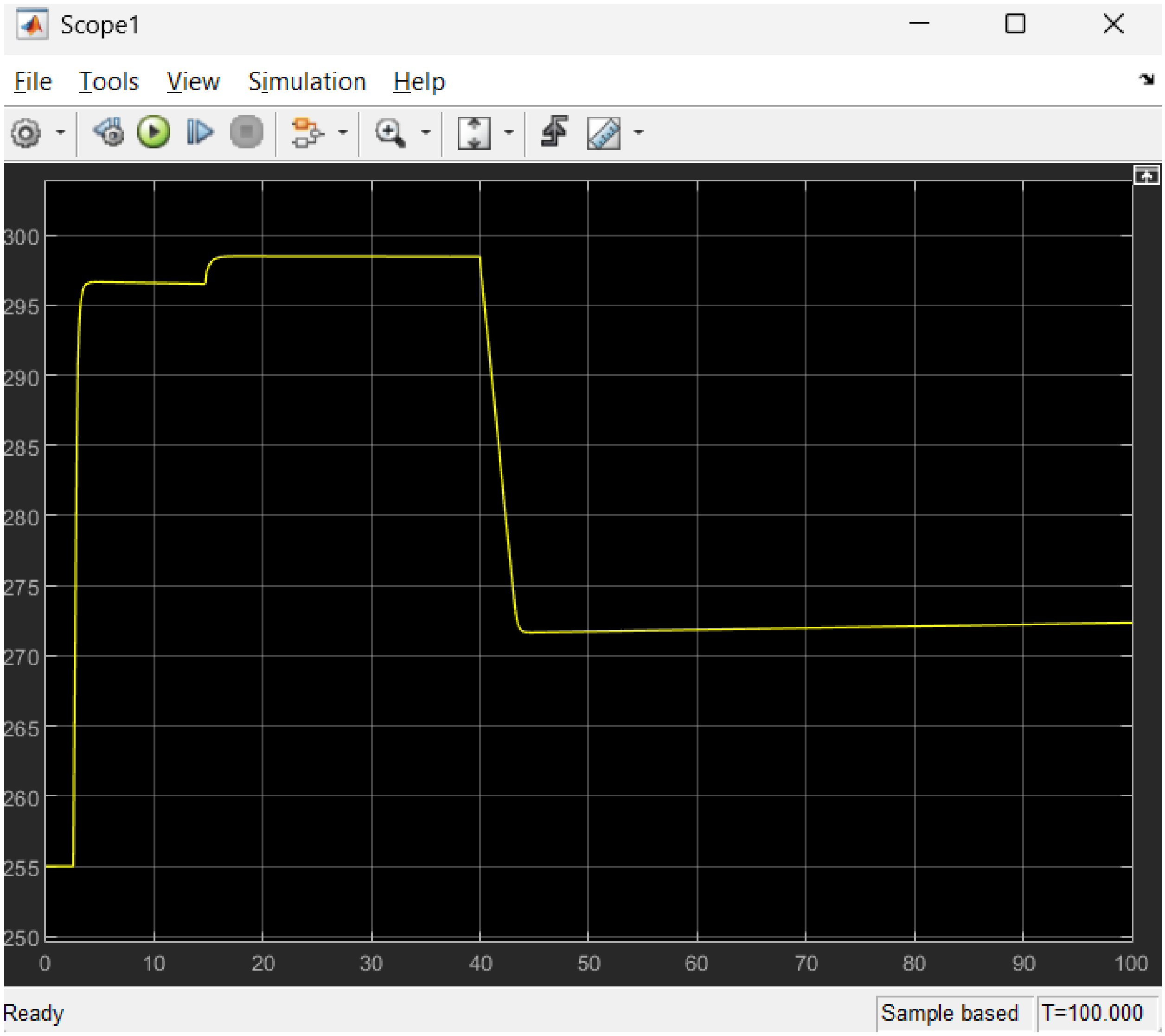

Figure (3)

As shown in Figure (3) the temperature rises from 255 K to 296K to go from 50 rad to 5.5 rad afterward rises again to 297 K to go 2.5 rad and stays at 297 for 25 seconds then starts decreasing to 273 to reach 6 rad before we show the martensite phase fraction plot and the curvature plot there is a big delay at first heating step and delay at the cooling step and it's obviously shown at cooling step 5 second after the input signal it reaches its 6 rad destination.

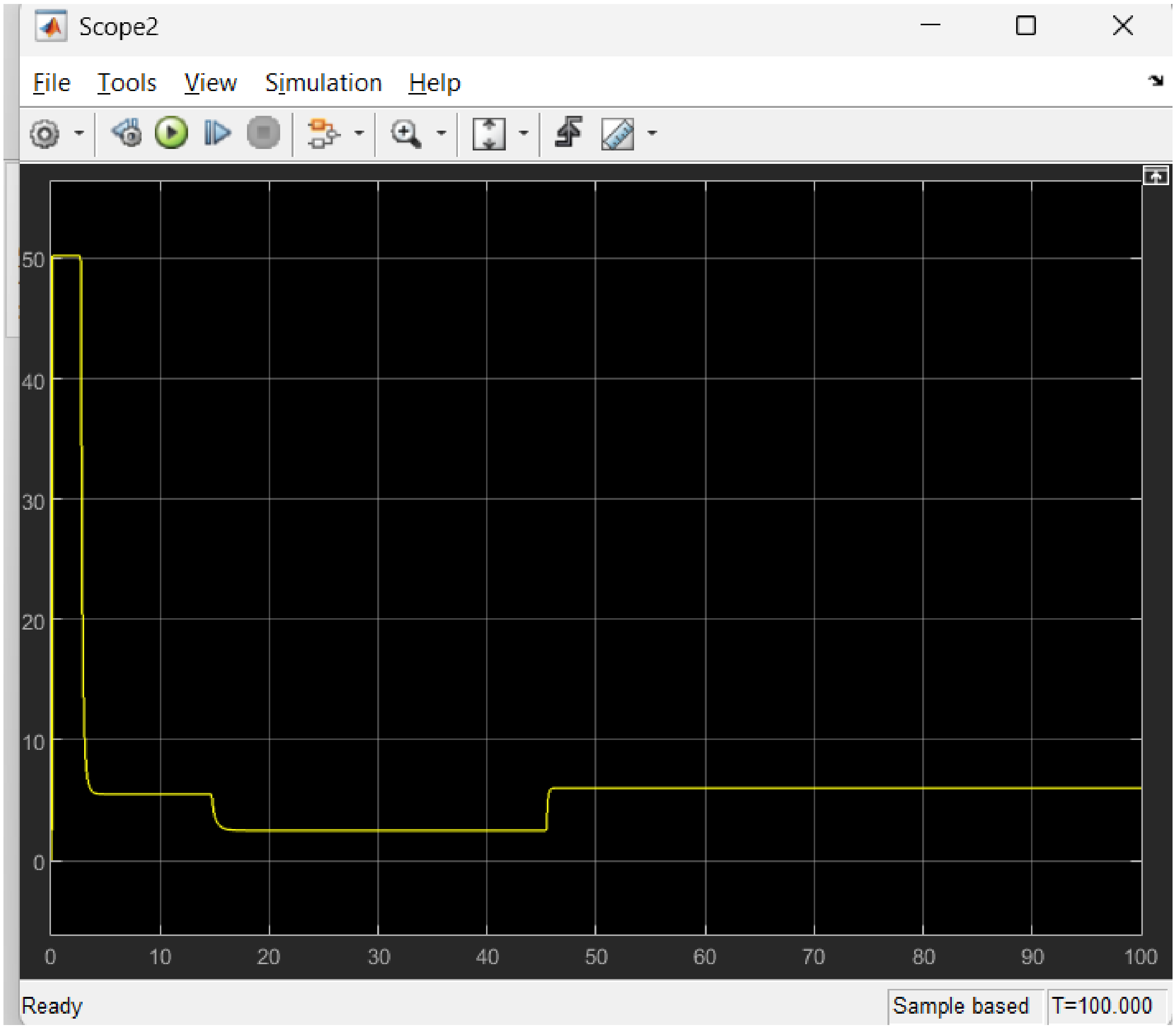

Figure (4) The curvature plot

As we have seen in the temperature plot it confirms the big delays at first destination step and the last one.

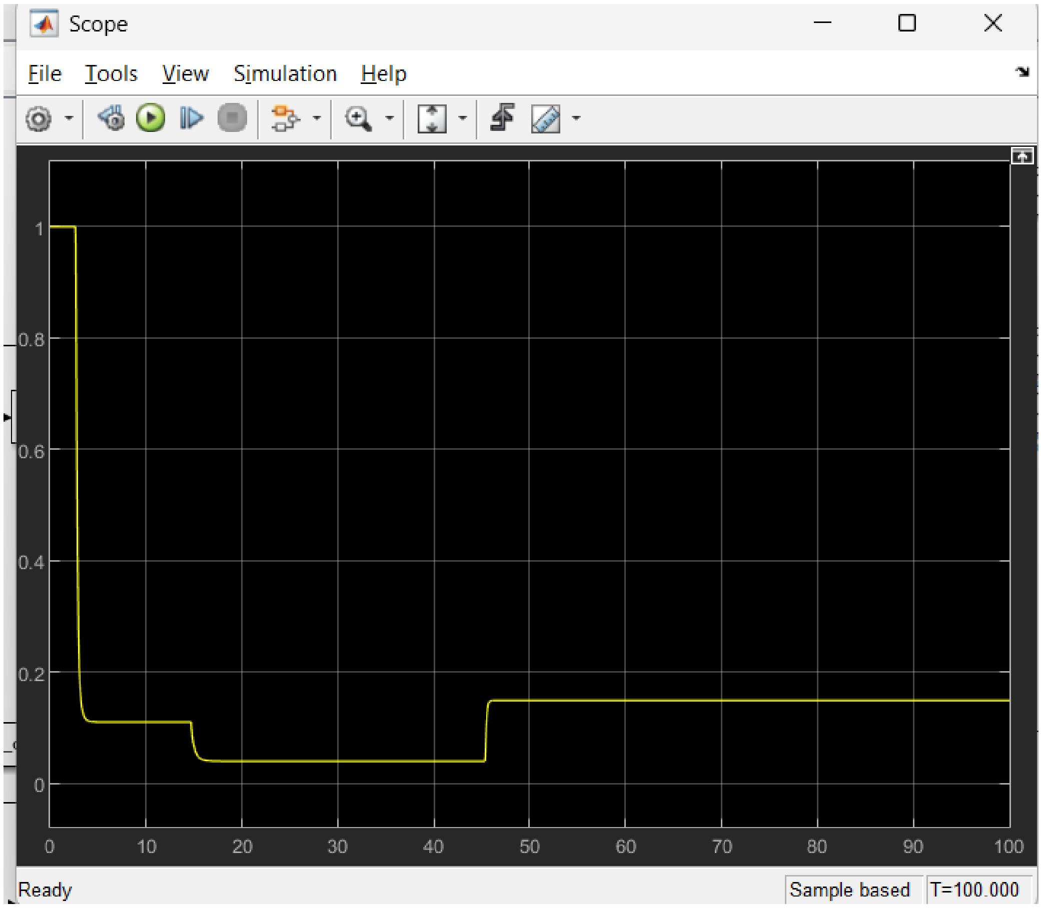

Figure (5) The martensite phase fraction plot

So I have a delay issue that cannot be treated with gains of the PI controller without neglecting real world physics like how fast the heat rises in the wire all this due to hysteresis of the wire.

4.Lookup table approach

Due to the delay problem that hysteresis is main cause of it and the main theory of PID controller I resort to feed forward controller that gives the wire the exact temperature it needs to go to its destination ,the main idea that I have a switch of cooling and heating path as example when the wire go from its maximum bending postion at 50 rad to 10 rad then it needs to go to 15 rad that heating and cooling path we should design an algorithm that does not confuse that there are two curvature values one in austenite phase and the another one in martensite phase .
First we need to feed our system a signal that covers all heating and cooling curve to design what is so called First Order Reversal Curves (FORC) ,the idea of it that I heat the

wire to go fully austensite and cool it to certain temperatures to cover all the wire's physics in my model I will take 50 samples from 320 K to 255 K

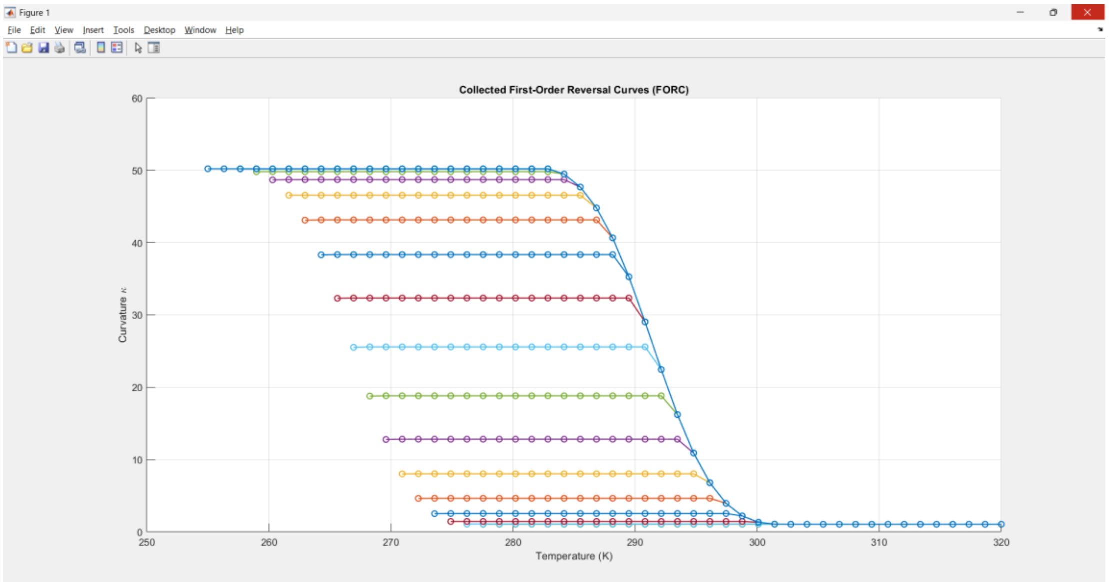

Figure (6) FORC

These curves represent how the system acts to different temperature values in heating and cooling paths then I will extract the two 1-D look up tables that are representing the feedforward controller.

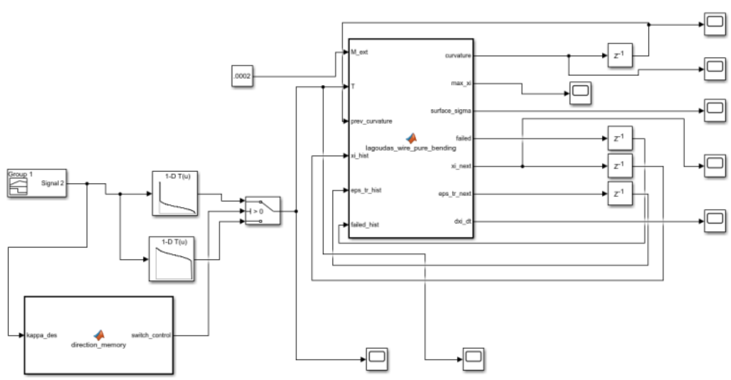

Figure (7) The final shape of the model

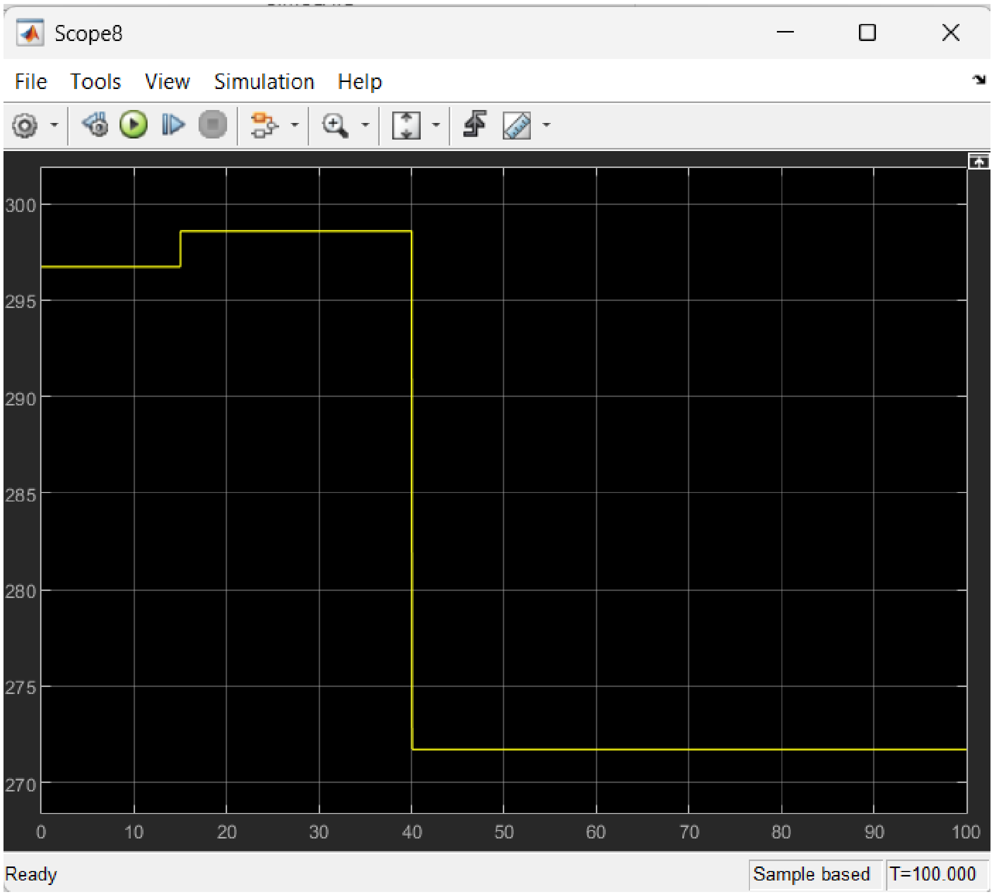

Figure (8) the input temperature to the system.

As shown the system targets the predefined temperature at which the desired curvature without delay (things will change if we use joule heating cause we can't jump to the targeted temperature instantly but it will still be faster if it knows where to go rather than the error calculation of the PID controller) .

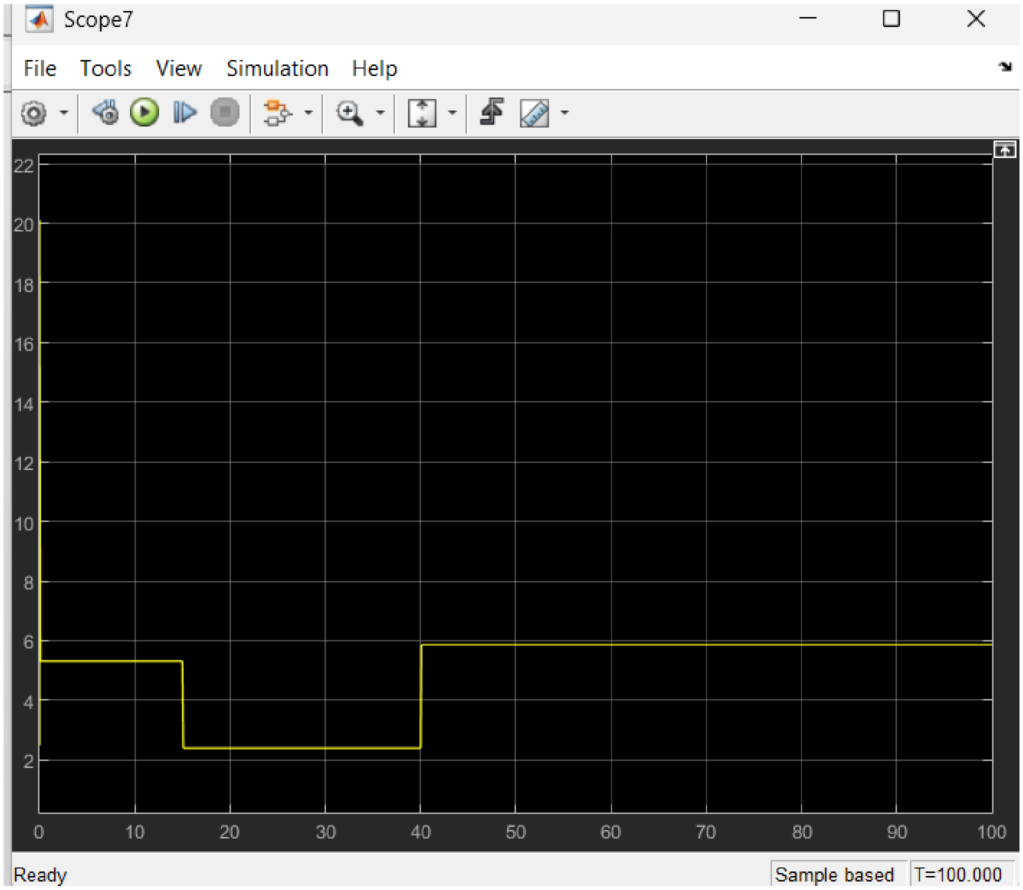

Figure (9) The output curvature

We see now the actual curvature and its nonrealistic tracking as we spoke before in real time or even modeling with joule heating it can't be like that.

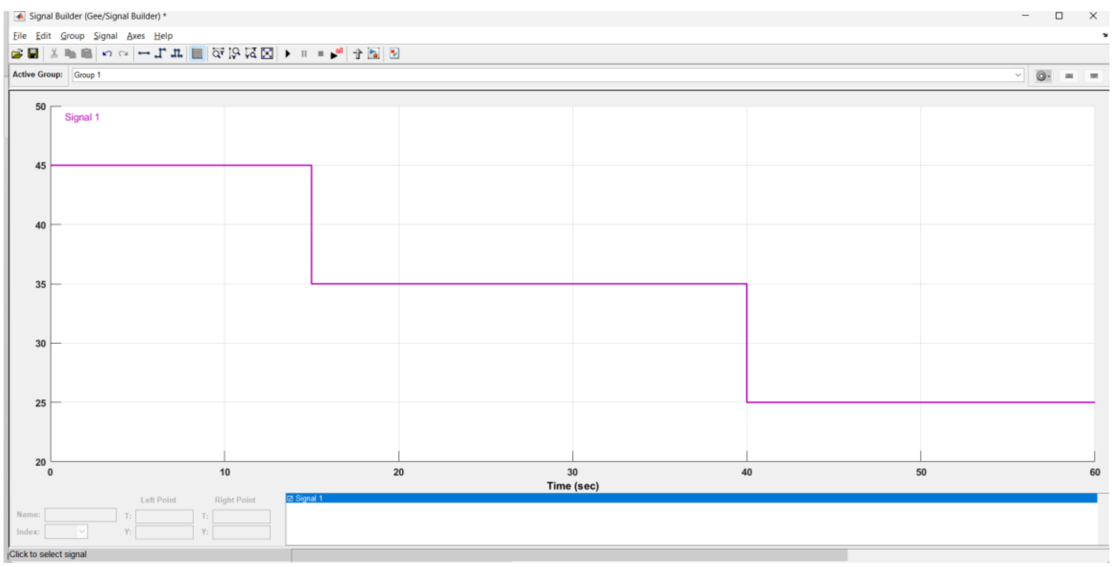

Figure (10) testing with another trajectory

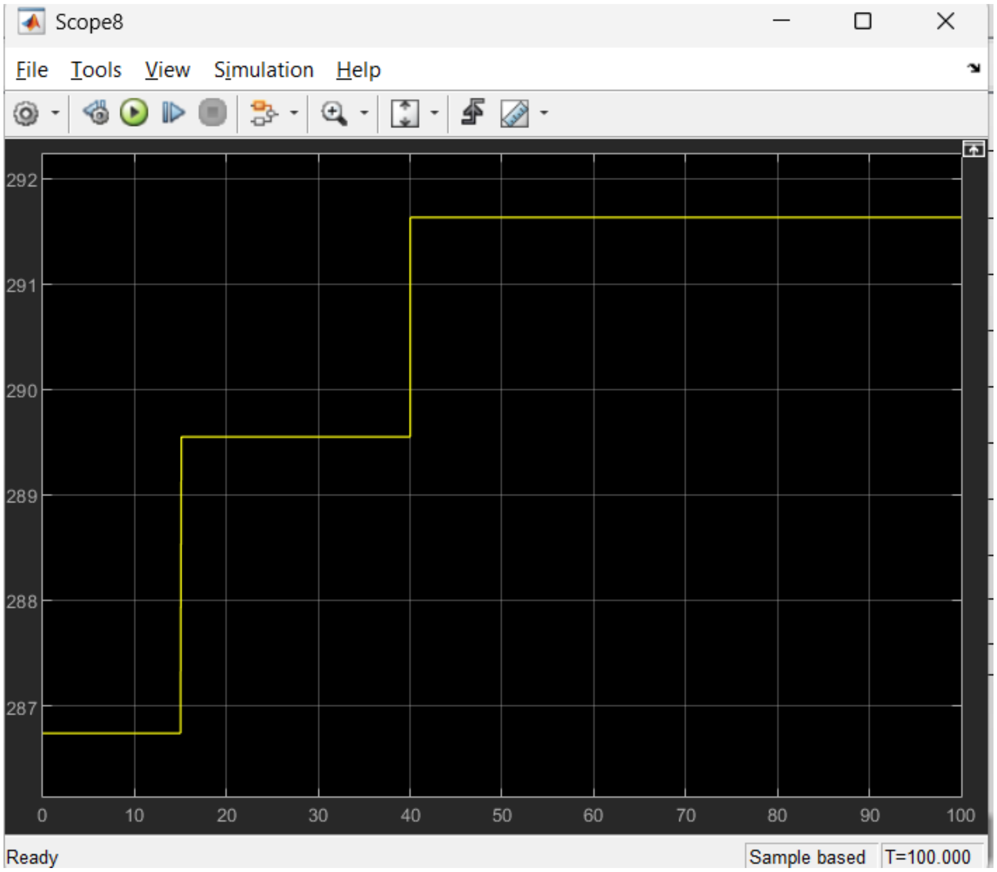

Figure (11) The input temperature plot of the new trajectory

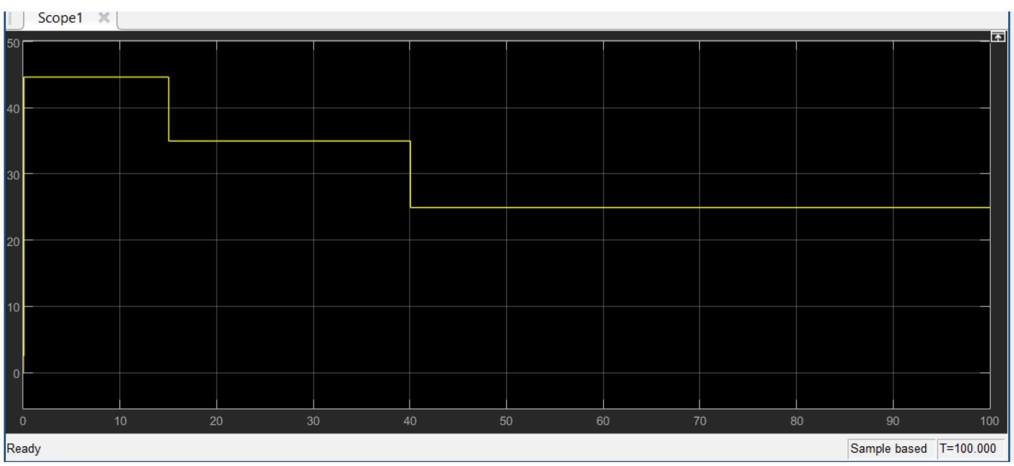

Figure (12) The output curvature of the new trajectory

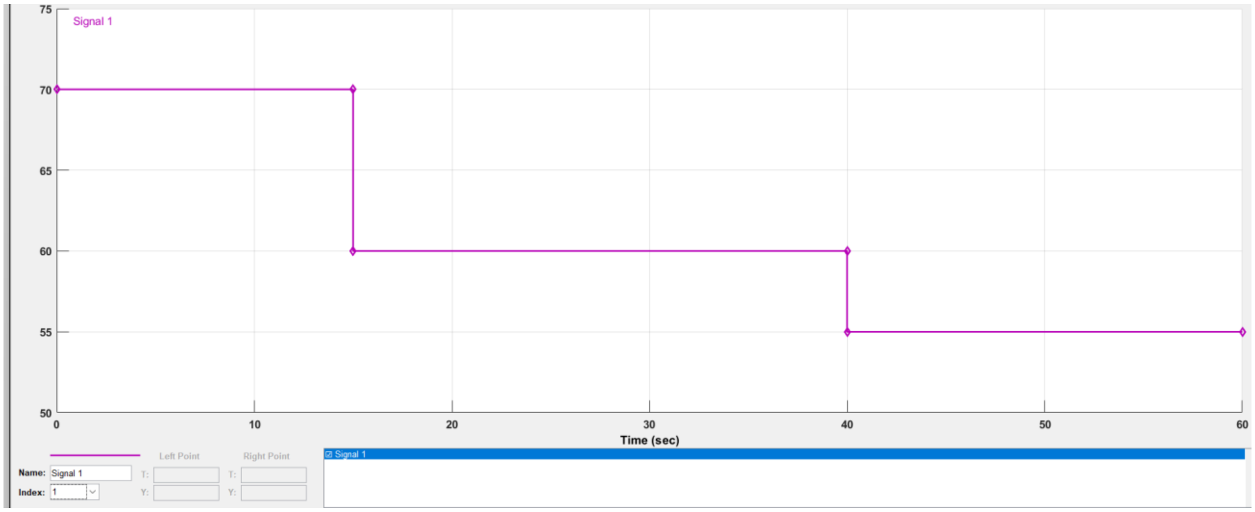

Figure (13) Another out of bounds trajectory signal

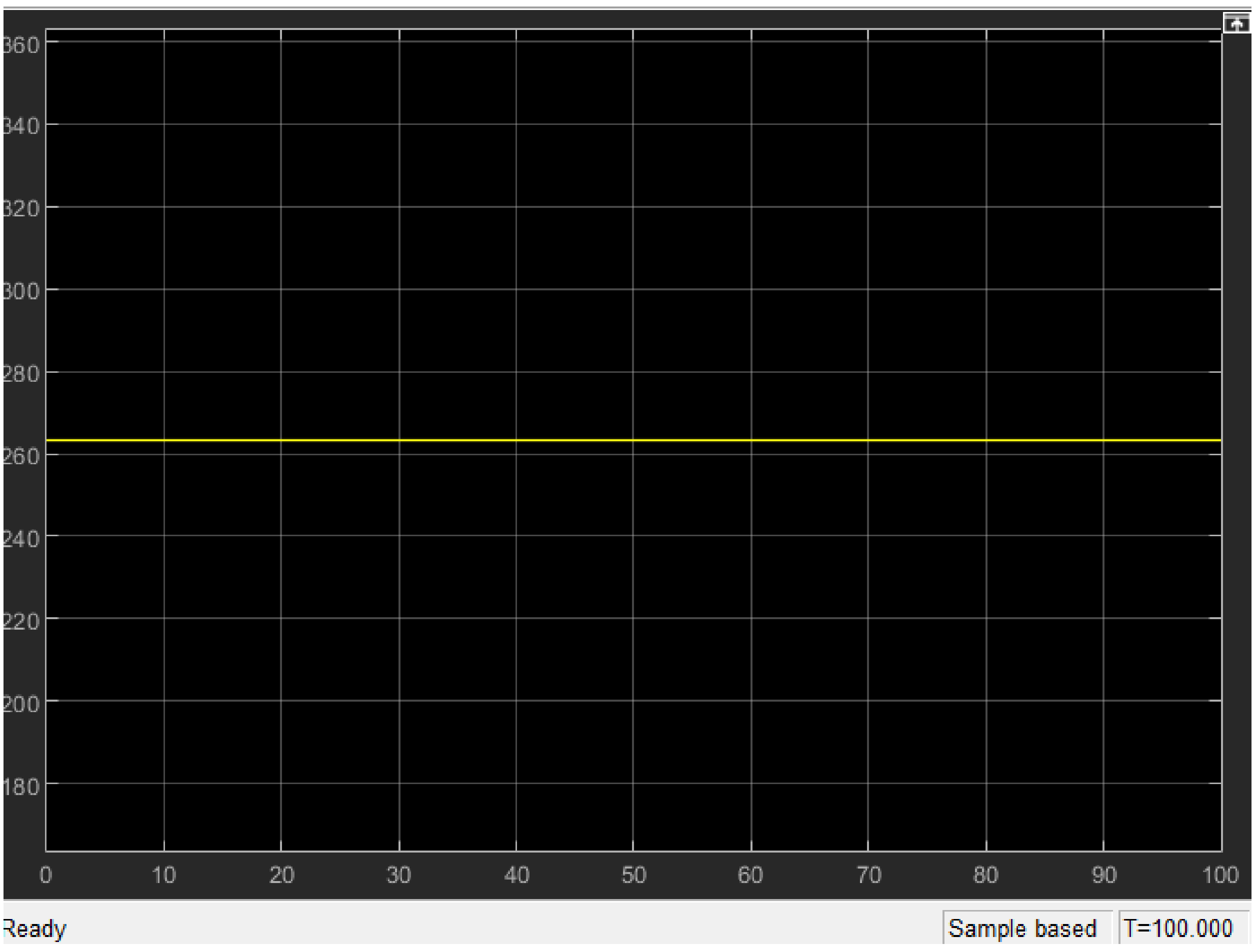

Figure (14) The unchanging temperature plot of the out of bound trajectory

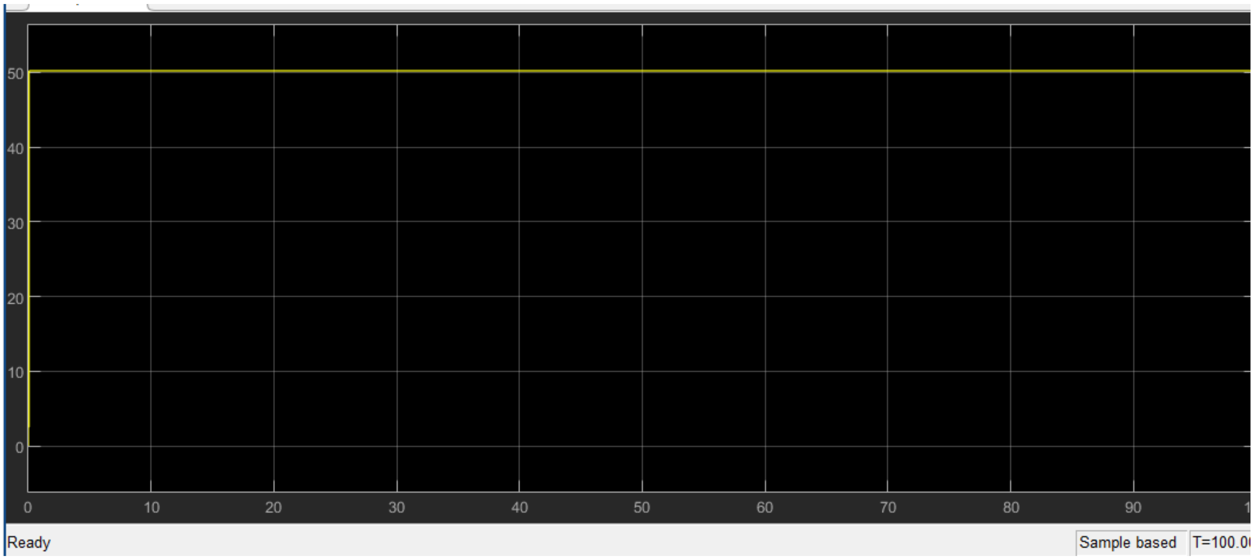

Figure (15) The wire hits its limit it can’t be bent more

5. The shortcomings of the model and how to deal with it in real time experimenting

If I change the load on wire I will be forced to do new FORC and extract another two lookup tables so it appears in real time I should get the FORC plot at the maximum transformation strain, I did not consider the heat transfer (Joule Heating) in my model although I knew there would be hysteresis in electric/heat part of the system too so I should do system identification to this part with power supply and thermocouple or thermal camera,

6. The conclusion

I discussed the hysteresis of the shape memory alloy and how it affects the control of SMA wire ,we have seen how the PID controller acts due to the hysteresis , then I have discussed a lookup table approach and how fast and accurate it is in comparison to the PID controller .